# Stick-together calculations

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Find a common final velocity using total momentum.

**SUCCESS CRITERIA - I can...**

1. Add the initial momenta with signs.
2. Use the combined mass after sticking.
3. Check the result against the direction of total momentum.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

Two carts separate from rest with no external horizontal impulse. One has momentum 15 kg·m/s left. Find the other's momentum.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $v_f = (m_1 v_{1,i} + m_2 v_{2,i})/(m_1 + m_2)$.
Sticking collision in one dimension with negligible external impulse. $v_f$ is the shared final velocity.

**Worked reasoning:** A 2 kg cart at +3 m/s sticks to a 1 kg cart at rest. Total momentum 6 kg·m/s divided by total mass 3 kg gives final velocity +2 m/s.

**Your turn:** Mark the conserved quantity on a before-and-after sketch. Explain why masses are added in the final denominator.

**Marzano check - circle:** ① ② ③

## 03 Look and explain

Two carts stick. Identify which masses belong in the final momentum expression and why.

Before contact: add velocity arrows.




**Marzano check - circle:** ① ② ③

## 04 Build the solution

A 1 kg cart moves right at 6 m/s and sticks to a 2 kg cart at rest. Find initial total momentum, combined mass and final velocity.

**Marzano check - circle:** ① ② ③

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### 05 Use the idea

A student averages the two starting velocities without considering mass. When would that happen to work?

**Marzano check - circle:** 1 2 3

### 06 Choose a method

A car follows a circle at steady 6 m/s. Sketch velocity at two points. Could net force be zero while it follows this path?

**Marzano check - circle:** 1 2 3

### 07 Exit check

Two equal 1 kg carts approach at 2 m/s in opposite directions and stick. Find final velocity.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

Choose the mass of a stationary cart so a 2 kg cart at 6 m/s slows to 2 m/s when they stick.

**Marzano check - circle:** 1 2 3